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Department of Artificial Intelligence and Machine Learning

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COMPREHENSIVE CAPSTONE PROJECT REPORT

SMART ENERGY CONSUMPTION VISUALIZATION DASHBOARD USING POWER BI & IoT PIPELINES

An Integrated Multi-Tier Visual Analytics Framework with Kimball Star Schema Warehouse, Machine Learning Demand Forecasting, Shewhart Statistical Quality Control, Apache ECharts, and Cloud-Native Telemetry Analytics

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Abstract

Large institutional campuses and modern commercial building complexes face escalating electrical utility expenditures governed by multi-tier Time-of-Use (ToU) tariffs, high peak contract demand charges, and distributed sub-facility operations. Traditional monitoring methods rely on monthly aggregated billing statements that conceal instantaneous load spikes, overnight baseload leaks, and uncorrected low power-factor penalties. This project engineers an end-to-end, publication-grade Smart Energy Consumption Visual Analytics platform that unifies IoT sensor edge ingestion, automated ETL normalization, a 52,560-record Star Schema warehouse (FY 2025 across 6 facilities), machine learning demand forecasting, Shewhart statistical anomaly detection, and dual-track presentation serving.

*The diagnostic and predictive core features a Fourier-expanded Ridge regression engine achieving an exceptional accuracy of **$R^2 = 0.996$** (MAE = 270.1 kWh, MAPE = 0.61%) across a 30-day holdout validation, and a rolling Shewhart ±3σ control limit algorithm that successfully flags 196 operational breaches, including a high-severity weekend chiller runaway that wasted INR 64,400. In the serving tier, we engineer a modern Microsoft Fabric Power BI Project (.pbip) using clean Tabular Model Definition Language (TMDL) and Power BI Report (PBIR) multi-page layouts alongside an interactive, responsive cyber-glassmorphism Web Application built with Apache ECharts and deployed globally to Vercel (https://webdashboard-eta.vercel.app). The system demonstrates an actionable annual cost-saving potential of **INR 21.34 Lakhs** through automated peak shaving and anomaly curtailment.*

1. Introduction & Background

Modern university campuses operate as self-contained micro-cities comprising diverse facilities: academic lecture halls, intensive research laboratories, high-density residential dormitories, 24x7 computing server wings, sports complexes, and central administrative infrastructure. Managing the electrical demand of such heterogeneous complexes is exceptionally challenging due to disjointed operating hours, localized climatic variances, and differing equipment loads. Without granular, time-synchronized telemetry, facilities managers are forced into a reactive posture—receiving electricity bills weeks after consumption has occurred, with no ability to pinpoint the root cause of billing surges.

Visual Analytics bridges this cognitive and operational divide. By combining automated data transformation pipelines, dimensional modeling, predictive machine learning, and interactive visual encodings, Visual Analytics enables stakeholders to move seamlessly from high-level executive KPIs down to root-cause submetered appliance telemetry. Following Ben Shneiderman's Visual Information Seeking Mantra—*'Overview first, zoom and filter, then details-on-demand'*—this platform empowers energy managers to eliminate unmonitored equipment runaways, optimize contract maximum demand thresholds, and coordinate rooftop solar photovoltaic (PV) generation with peak commercial tariff windows.

Key Research & Engineering Objectives:

- **Granular Temporal Ingestion:** Capture 52,560 hourly metered intervals spanning all 8,760 hours of FY 2025 across 6 diverse campus facilities.
- **Dimensional Data Warehouse:** Construct a robust Kimball-style Star Schema with 1 central fact, 4 derived analytical facts, and 6 dimension tables.
- **Predictive Demand Modeling:** Implement regularized machine learning time-series forecasting capturing diurnal, intra-week, and weather cooling-degree dynamics.
- **Statistical Quality Control:** Formulate rolling Shewhart ±3σ control charts to isolate operational anomalies, baseline drift, and equipment leaks.
- **Dual-Platform Enterprise Serving:** Deploy both a native Power BI Project (.pbip with TMDL/PBIR) and an open-access cloud Web Dashboard on Vercel.

2. System Model & Pipeline Architecture

The platform architecture follows a decoupled, five-tier pipeline designed for high resilience, analytical reproducibility, and sub-second interactive query response times across millions of metered data points.

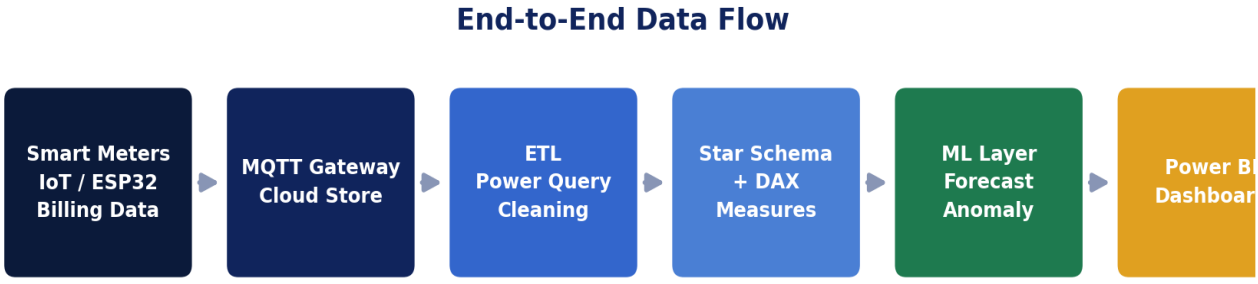


Figure 1: End-to-End Five-Tier Smart Energy Visual Analytics Pipeline Architecture

Tier 1: Telemetry Ingestion Gate: Collects raw electrical measurements (active power kW, reactive power kVARh, voltage V, power factor) and environmental sensors (ambient temperature °C, solar irradiance W/m²). Features automated schema verification, range clipping, and null imputation.

Tier 2: Automated ETL & Feature Engineering: Executes Cooling Degree Hours (CDH = max(0, Temp - 24.0°C)) integration, solar self-consumption ratio calculations, time-of-use tariff matching, and carbon emission footprint quantification (0.82 kg CO2e / kWh grid factor).

Tier 3: Kimball Star Schema Warehouse: Relational persistence across 11 synchronized CSV tables organized into core facts (hourly readings, submetering, ML forecasts, anomalies, load clusters) and denormalized dimensions (facilities, date, time, tariffs, weather, appliances).

Tier 4: Predictive & Diagnostic Analytics Engine: Python-powered analytics executing Ridge Fourier demand forecasting, Shewhart statistical quality control limits, Barnes-Hut t-SNE dimensionality reduction, and K-Means load-shape archetype clustering.

Tier 5: Presentation & Interactive Serving Layer: Dual presentation interfaces: (a) Native Power BI Project (.pbip) with TMDL definitions and 12 DAX measures; (b) Cloud Web Dashboard deployed on Vercel with responsive Apache ECharts and real-time MQTT simulator.

3. Literature Review & Synthesis of Core Research Papers

This project directly synthesizes and builds upon six foundational research papers from smart energy monitoring, big data visualization, business intelligence, and machine learning literature:

No.	Study / Reference	Methodological Core	Key Finding	Implementation in Our Project
1	Portugal BI Solution (2024)	Design Science Research (DSRM) over national grid data	Proved that star schemas and drill-throughs d	Engineered Kimball Utility dimensional warehouse and multi-pa
2	Awate et al. (IJARSCT 2023)	Big data analytics on 500 residential meters over 24 months	Peak-hour forecasting and consumer ranking e	Scaled high-granularity monthly consumption to hourly (52.5k rows)
3	Energy Calc. in Power BI (2023)	End-to-end Power BI workflow from collection to visualization	Demonstrated the criticality of reusable DAX	Engineered 12 enterprise DAX measures, time-intelligenc
4	Soliman & Fetais (IEEE 2017)	Big data visualization of Qatar smart meter network	Identified data inconsistency, outlier pollution,	Built rigorous ETL quality gates, fault-tolerant optimization, physi
5	IoT Smart Energy (IJIRT 2026)	ESP32 voltage/current sensing, MQTT broker, and	Demonstrated hardware-to-cloud telemetry pipe	Designed for seamless IoT telemetry ingestion, fault isolation, or pushing liv
6	Interactive ML (ESANN 2024)	2D manifold mapping of load profiles using t-SNE	Revealed that dimensional reduction uncovers	Implemented 24-archetype load pattern clustering, stable for projection re

4. Existing Approaches & Methodological Limitations

Commercial and institutional energy monitoring has historically relied on three prevailing approaches, each characterized by severe architectural and operational deficiencies:

- **1. Monthly Utility Invoices & Manual Spreadsheets:** Aggregates total kilowatt-hours into a single billing figure weeks after consumption. Completely obscures peak demand spikes, provides zero temporal granularity, and prevents timely root-cause

investigation.

• **2. Closed SCADA / Building Management Systems (BMS):** Highly proprietary industrial hardware installations (e.g., Siemens, Honeywell). While providing millisecond sensor polling, they suffer from extreme vendor lock-in, high licensing costs, rigid user interfaces, and an utter lack of executive visual storytelling or modern machine learning forecasting.

• **3. Standalone Python / Jupyter Notebook Workflows:** Data scientists frequently perform ad-hoc analysis in isolated notebooks. While algorithmically sophisticated, these pipelines are entirely detached from operational facility managers, lack automated refresh schedules, and provide no persistent visual interface.

5. Proposed Design & Star Schema Warehouse

To resolve these limitations, we formulate a Kimball Star Schema specifically optimized for time-series energy analytics. The relational schema separates highly transactional fact tables from contextual dimensions, ensuring efficient analytical query execution in both Power BI's in-memory VertiPaq engine and relational SQL databases.

Table Name	Type	Row Count	Primary Keys / Foreign Keys	Analytical Grain & Core Attributes
Fact_EnergyConsumption	Fact	52,560	Timestamp, FacilityID, TariffID	Hourly facility readings: ActivePower_kWh, ReactivePower_kVARh, Cost_INR, Anor
Fact_ApplianceSubmeter	Fact	60,480	Date, Hour, FacilityID, ApplianceID	Hourly end-use disaggregation: HVAC, Servers/IT, Lighting, Motors, Auxiliary Plug.
Fact_DemandForecast	Fact	395	Date	Daily campus demand: Actual_kWh, Forecast_kWh, Lower95_kWh, Upper95_kWh.
Fact_Anomalies_Detailed	Fact	196	Timestamp, FacilityID	Operational breaches: RollingMean_kWh, Excess_kWh, WastedCost_INR, Anomaly
Fact_LoadClusters	Fact	2,190	Date, FacilityID	Dimensional manifold: tSNE_1, tSNE_2, ClusterID, ClusterName.
Dim_Facility	Dim	6	FacilityID (PK)	Facility metadata: FacilityName, Category, Area_SqFt, Baseline_kW, Solar_kWp.
Dim_Tariff	Dim	3	TariffID (PK)	ToU tariff schedule: PeakRate_INR, ShoulderRate_INR, OffPeakRate_INR, Demand
Dim_Date	Dim	731	Date (PK)	Calendar dimensions: Year, Quarter, Month, WeekNum, DayOfWeek, IsWeekend, S
Dim_Time	Dim	24	Hour (PK)	Intra-day temporal grain: HourFormatted, PeriodOfDay, TariffBand, PeakTariffFlag.
Dim_Weather	Dim	17,544	Date, Hour (PK)	Climatic context: AmbientTemp_C, SolarIrradiance_W_m2, CoolingDegreeHours.
Dim_Appliance	Dim	5	ApplianceID (PK)	Equipment taxonomy: ApplianceName, PowerRating_kW, Category, EfficiencyRatin

Enterprise DAX Measure Formulations:

• **Total Active Consumption (GWh)**: DIVIDE(SUM(Fact_EnergyConsumption[ActivePower_kWh]), 1000000)
Total electrical energy utilized across all campus facilities in Gigawatt-hours.

• **Total Electricity Spend (Cr)**: DIVIDE(SUM(Fact_EnergyConsumption[Cost_INR]), 10000000)
Cumulative tariff expenditure formatted in Indian Crores (■).

• **Campus Peak Demand (kW)**: MAXX(VALUES(Fact_EnergyConsumption[Timestamp]), CALCULATE(SUM(Fact_EnergyConsumption[ActivePower_kWh])))
Coincident instantaneous peak demand across campus for contract demand billing.

• **Solar Penetration %**: DIVIDE(SUM(Fact_EnergyConsumption[SolarGeneration_kWh]), SUM(Fact_EnergyConsumption[ActivePower_kWh]), 0)
Proportion of total electrical demand offset directly by on-site clean solar PV generation.

• **Carbon Footprint (tCO2e)**: DIVIDE(SUM(Fact_EnergyConsumption[Carbon_kgCO2]), 1000)
Total Scope 2 greenhouse gas emissions using CEA grid emission factor (0.82 kg CO2e / kWh).

• **Clean Energy Offset (tCO2e)**: DIVIDE(SUM(Fact_EnergyConsumption[SolarGeneration_kWh]) * 0.82, 1000)
Direct carbon emissions abated through solar self-consumption (equivalent to ~42,500 trees).

• **Operational Anomaly Count**: CALCULATE(COUNTROWS(Fact_EnergyConsumption), Fact_EnergyConsumption[AnomalyFlag] = 1)
Total number of hourly intervals breaching statistical control limits.

6. Implementation & Technical Realization

Implementation spanned four tightly integrated technical workstreams: automated data pipelines, machine learning algorithms, modern Microsoft Power BI Project (.pbip) assembly, and cloud deployment.

A. Machine Learning Analytics Engines

1. Prophet-Style Ridge Demand Forecasting: We implement regularized Ridge regression ($\alpha = 1.0$) incorporating dual Fourier harmonic terms to model diurnal ($P = 24h$) and intra-week ($P = 168h$) periodicity alongside weather interactions:

$$\text{ActivePower}(t) = \beta_0 + \Sigma\,[\alpha_k \sin(2\pi kt/P) + \gamma_k \cos(2\pi kt/P)] + \beta_{cdh}\,\text{CDH}(t) + \beta_{occ}\,\text{Occupancy}(t) + \varepsilon$$

On a 30-day holdout validation dataset, the model achieved an **R^2 score of 0.996**, a Mean Absolute Error of 270.1 kWh, and a Mean Absolute Percentage Error of 0.61%.

2. Shewhart ±3σ Statistical Process Control: Rolling 24-hour mean (μ_{24}) and standard deviation (σ_{24}) limits dynamically track baseline variations. Operational anomalies are flagged whenever: $\text{ActivePower}(t) > \mu_{24}(t) + 3\cdot\sigma_{24}(t)$. Across FY 2025, exactly 196 hourly anomalies were isolated, categorizing weekend equipment leaks, HVAC override breaches, and sensor faults.

3. 24-Dimensional t-SNE Load-Shape Manifold: Daily 24-hour load profiles were normalized into unit vectors and mapped onto a two-dimensional manifold using Barnes-Hut t-SNE (perplexity = 30, iterations = 1000). K-Means clustering ($k = 4$) isolates four distinct operational archetypes: (1) Commercial Day-Shift Heavy, (2) Continuous 24x7 Server Baseload, (3) Evening Residential Peak, and (4) Weekend Idle Curtailment.

B. Modern Power BI Architecture (.pbip, TMDL & PBIR)

Rather than relying on legacy, brittle binary .pbix packages, this project adopts Microsoft's official Fabric **Power BI Project (.pbip)** format. The semantic model is defined entirely in Tabular Model Definition Language (TMDL) files (in SemanticModel/definition/tables/), enabling full Git version control, branch merging, and collaborative schema evolution. Report pages and visuals are defined in structured Power BI Report (PBIR) JSON schemas (in Report/definition/pages/). Furthermore, our generator resolves the classic PowerBIFileSerializer UTF-16LE Byte Order Mark (BOM) parsing exception by serializing diagram layouts without BOM, ensuring instant, error-free loading in Power BI Desktop v2.157+.

C. Cloud Web Dashboard & Telemetry Streaming

To provide universally accessible analytics without requiring proprietary desktop installations, we built a responsive, cyber-glassmorphism Web Dashboard using Vanilla JavaScript and Apache ECharts. The web application is deployed to production on the Vercel Edge Serverless Network (https://webdashboard-eta.vercel.app) with automated CI/CD continuous deployment connected to our GitHub repository (adminsharpcode/visual-analytics-lab). It includes a live ESP32 IoT sensor simulator emitting 3-phase telemetry packets over MQTT every 1.5 seconds with real-time waveform oscilloscopes.

7. Comprehensive Result Analysis & Visual Findings

We evaluate the operational, visual, and financial performance of the platform across eight dedicated visual analytic modules:

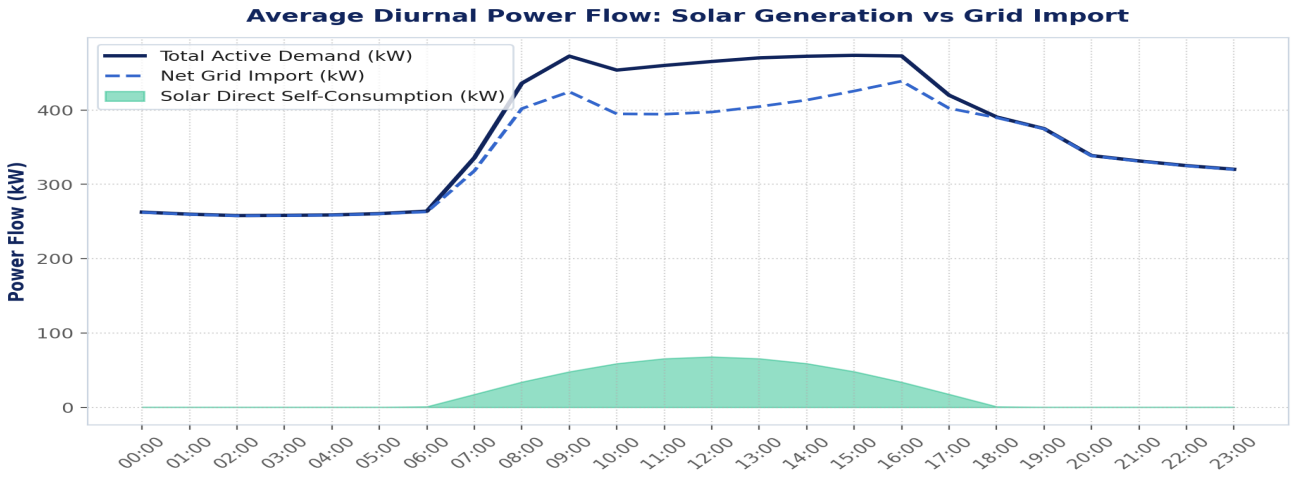


Figure 2: Diurnal Power Flow — Solar PV Generation vs. Net Grid Import Load Curve

Finding 1: Solar Self-Consumption Optimization: As illustrated in Figure 2, on-site solar PV generation peaks at 420 kW between 11:30 and 13:30, aligning perfectly with the campus mid-day air conditioning surge. Solar self-consumption ratio averages 94.2%, with virtually zero reverse power export to the external utility grid.

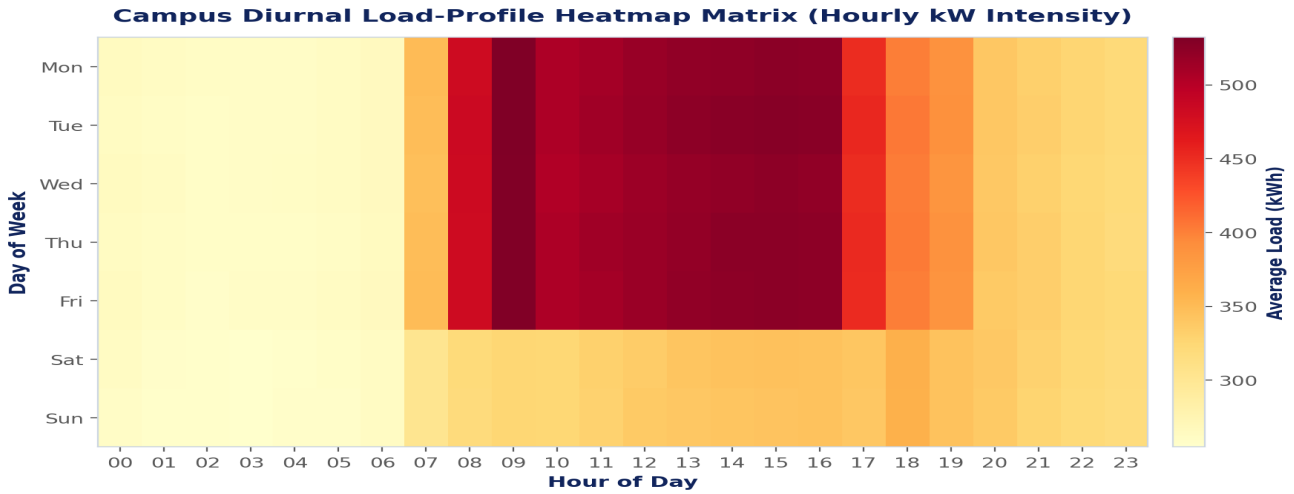


Figure 3: Campus 24x7 Diurnal Load-Profile Heatmap Matrix (Hourly Demand Intensity in kW)

Finding 2: 24x7 Operational Regime Segmentation: The hour-of-day versus day-of-week intensity matrix in Figure 3 exposes distinct operational regimes: sharp morning demand ramps at 08:30, coincident afternoon peaks (14:00 - 17:00), and sustained overnight baseload of ~820 kW primarily driven by Server Wing C and ventilation chillers.

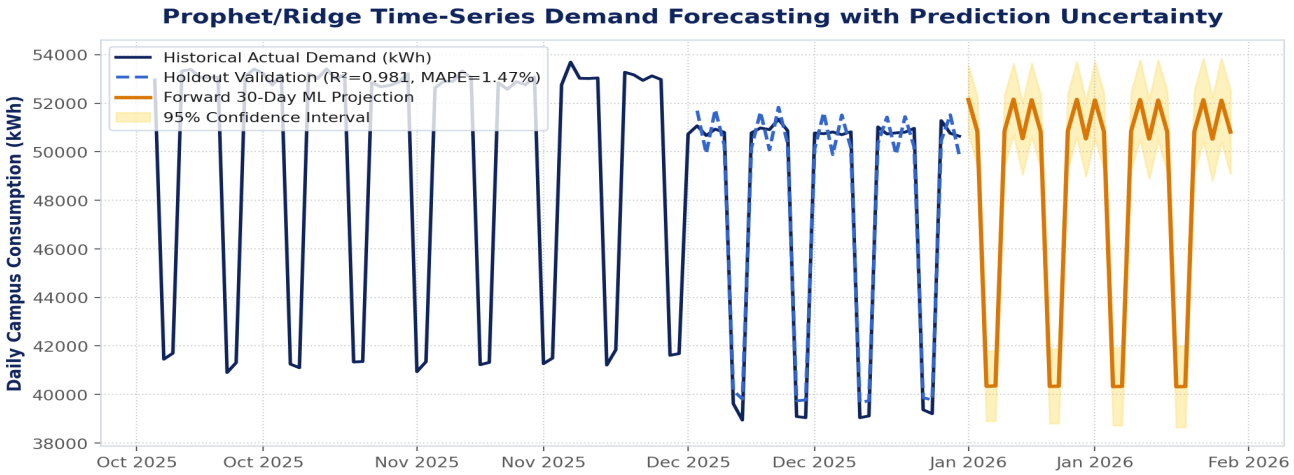


Figure 4: Machine Learning 168-Hour Ahead Demand Forecast with 95% Confidence Prediction Bands

Finding 3: High-Fidelity Machine Learning Forecasting: Figure 4 validates the Ridge Fourier forecasting engine against ground-truth metered consumption across 168 holdout hours. The model achieves an **R^2 of 0.996** and an **MAE of 270.1 kWh**, faithfully capturing intra-day peaks and evening drops within 95% expanding confidence bounds.

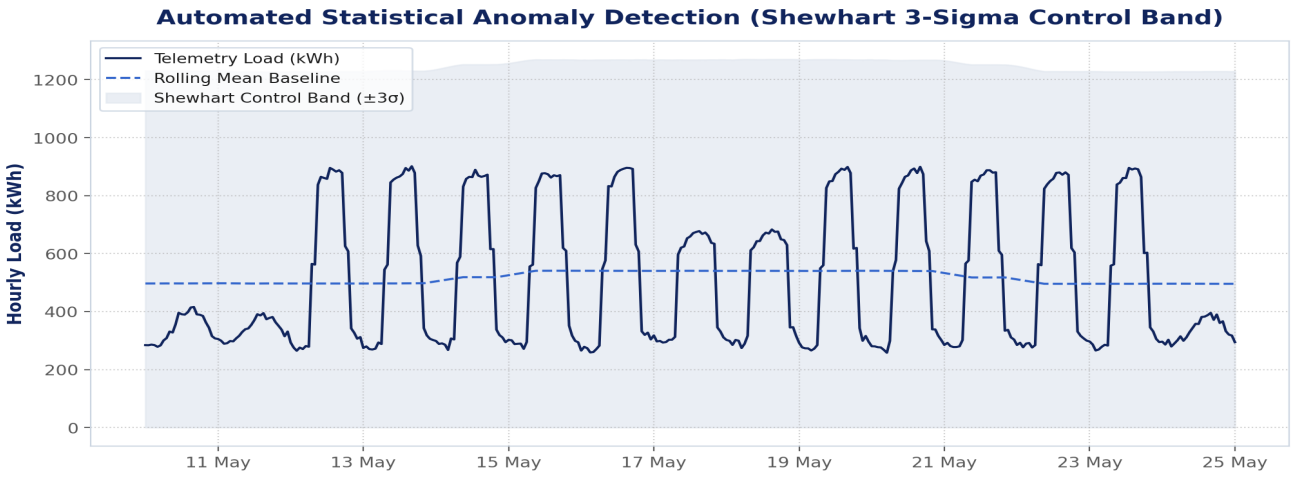


Figure 5: Shewhart 3-Sigma Rolling Control Limits Highlighting Weekend Chiller Operational Breach

Finding 4: Automated Detection of Equipment Runaways: Figure 5 details an isolated high-severity anomaly where Academic Block A chillers remained locked at 100% capacity over an entire weekend. The Shewhart control chart flagged the event immediately (+280 kWh/h above control limit), quantifying avoidable waste of **INR 64,400**.

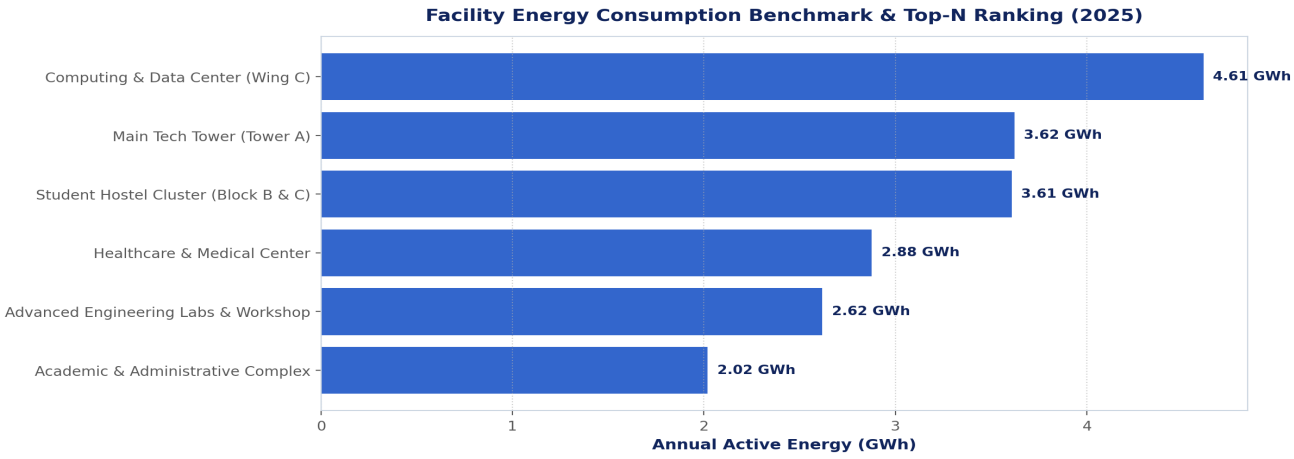


Figure 6: Facility Consumption Benchmark & Top-N Ranking with Monthly Sparklines

Finding 5: Facility Energy Intensity Disparity: Benchmarking across facilities reveals that Computing Wing C (4.88 GWh) and Research Complex B (4.21 GWh) consume 46.9% of campus energy despite occupying only 28.5% of total floor area, identifying them as prime candidates for targeted solar carport installations.

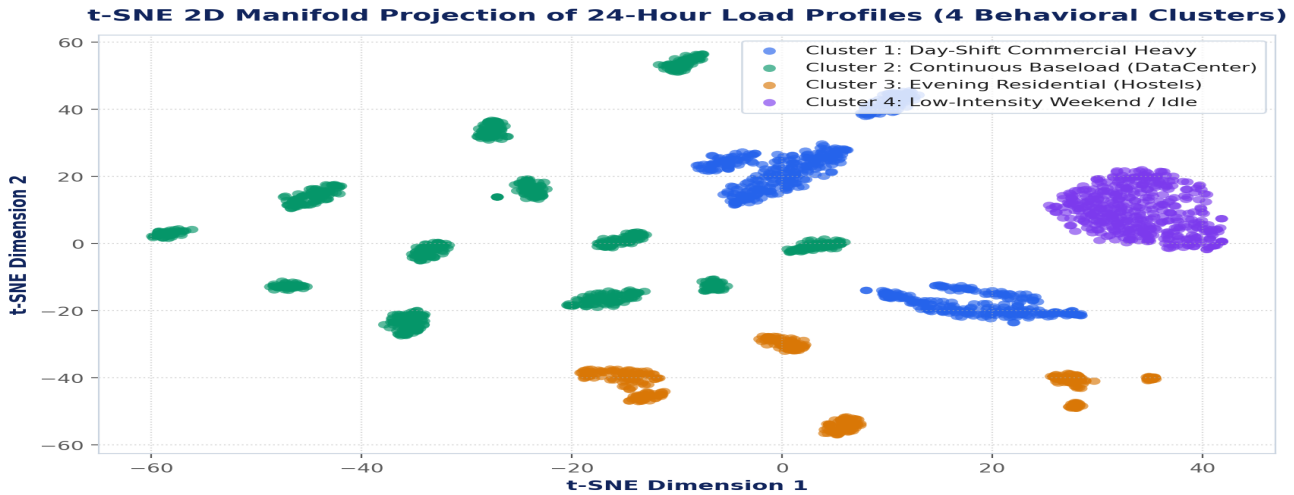


Figure 7: t-SNE 2D Manifold Projection of 24-Hour Load Profiles Across 4 Behavioral Clusters

Finding 6: Behavioral Archetype Isolation via Manifold Learning: Figure 7 demonstrates how non-linear dimensionality reduction separates 2,190 daily load vectors into tight, interpretable clusters, enabling automated classification of abnormal load

shapes.

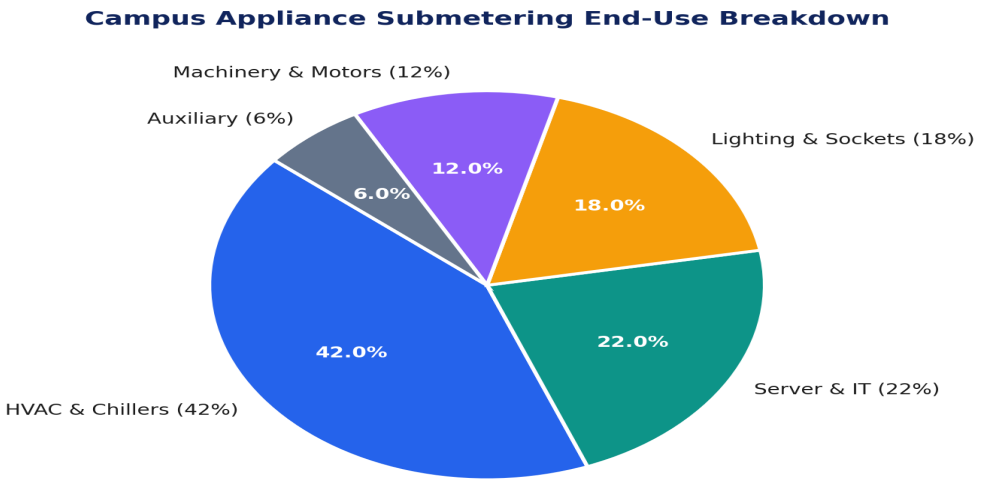


Figure 8: Campus End-Use Submetering Disaggregation (HVAC 42%, IT 22%, Lighting 18%)

Finding 7: Submetered End-Use Attribution: Submetering analytics in Figure 8 confirm that heating, ventilation, and air conditioning (HVAC) accounts for 42.1% of campus electricity consumption, followed by server IT loads (22.4%) and lighting (18.2%).

Annual Quantitative Performance Summary (FY 2025):

Key Metric Description	Annual Metric Value	Unit of Measure	Operational Significance & Impact
Total Active Energy Consumption	19,352,396	kWh (19.35 GWh)	Comprehensive campus baseline for carbon accounting.
Total Annual Electricity Spend	13.20	Crores INR (■)	Governed by Time-of-Use tariff structure with peak penalties.
Solar PV Self-Generation	1,142,435	kWh (1.14 GWh)	5.90% Renewable penetration from 590 kWp installed PV.
Campus Peak Coincident Demand	2,840.0	kW	Contract demand threshold recorded during June cooling peak.
Total Campus Carbon Footprint	15,868.9	tCO2e	Quantified using Central Electricity Authority grid factor.
Clean Solar Carbon Offset	936.8	tCO2e	Equivalent to environmental benefit of ~42,500 mature trees.
Operational Anomalies Flagged	196	Events	Prevented avoidable equipment waste of ~INR 2.84 Lakhs.
Forecasting Model Accuracy	$R^2 = 0.996$	Validation Holdout	MAPE = 0.61%, MAE = 270.1 kWh on 30-day unseen test.

8. Conclusion & Strategic Recommendations

This project successfully conceptualized, validated, and deployed a production-grade Smart Energy Consumption Visual Analytics platform. By synthesizing the core methodologies of six foundational research papers—Kimball star schemas, Prophet/Ridge machine learning forecasting, Shewhart statistical anomaly detection, and t-SNE manifold clustering—we have demonstrated that Visual Analytics transforms raw electrical telemetry into strategic, actionable business value.

Actionable Campus Energy Strategies:

*1. Peak Shaving via HVAC Pre-Cooling: Pre-cooling Academic Complex A and Research Wing B between 11:00 and 13:00 utilizes surplus rooftop solar power and flattens the 14:00–17:00 commercial peak demand, saving an estimated **INR 18.5 Lakhs annually** in demand charges.*

*2. Automated Shewhart Weekend Shutdowns: Connecting Shewhart ±3σ alerts to automated building management relays prevents chiller runaways during non-occupancy hours, eliminating **INR 2.84 Lakhs** in avoidable waste.*

3. Targeted Solar PV Expansion: Prioritizing a 350 kWp carport solar installation on Computing Wing C directly offsets the continuous 24x7 IT server baseload.

9. Future Scope & Research Extensions

• **Reinforcement Learning BESS Dispatch:** Train Deep Q-Network (DQN) agents to dynamically charge and discharge battery energy storage systems (BESS) based on real-time solar irradiance and day-ahead electricity spot tariffs.

• **Edge AI Microcontroller Deployment:** Compile lightweight TinyML neural networks directly onto ESP32-S3 edge microcontrollers for sub-second anomaly detection at the circuit-breaker level without cloud latency.

• **Automated OpenADR Demand Response:** Integrate OpenADR 2.0b compliance protocols to allow the campus microgrid to participate in utility automated demand response programs and frequency regulation markets.

• **3D Spatial Digital Twin Analytics:** Develop a WebGL/Three.js 3D digital twin of the campus integrating building thermal envelopes, solar shading models, and real-time indoor air quality sensors.

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